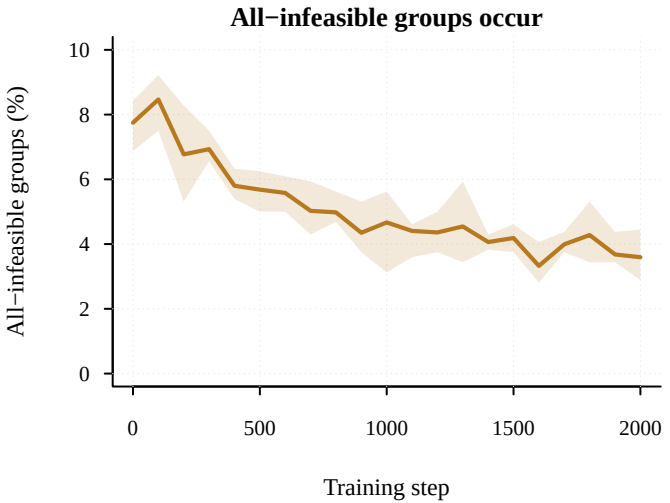
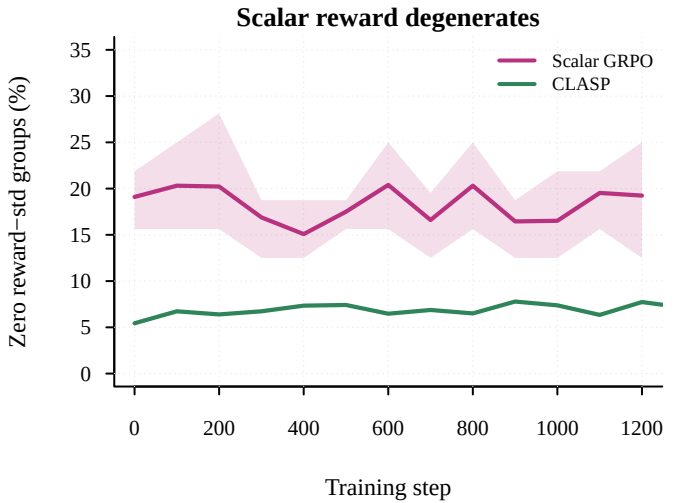
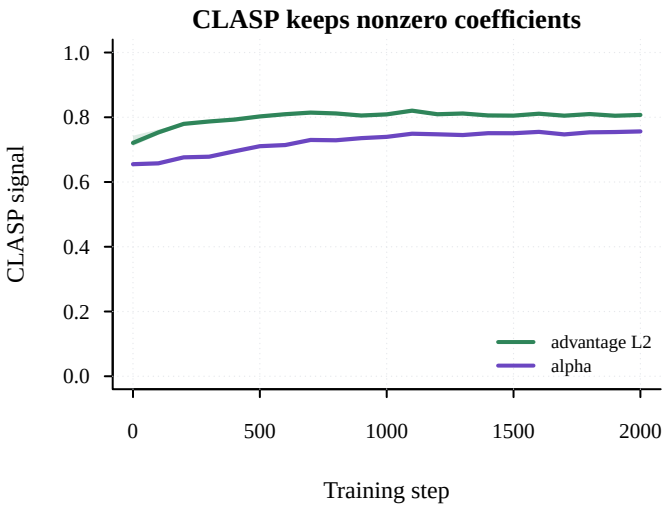
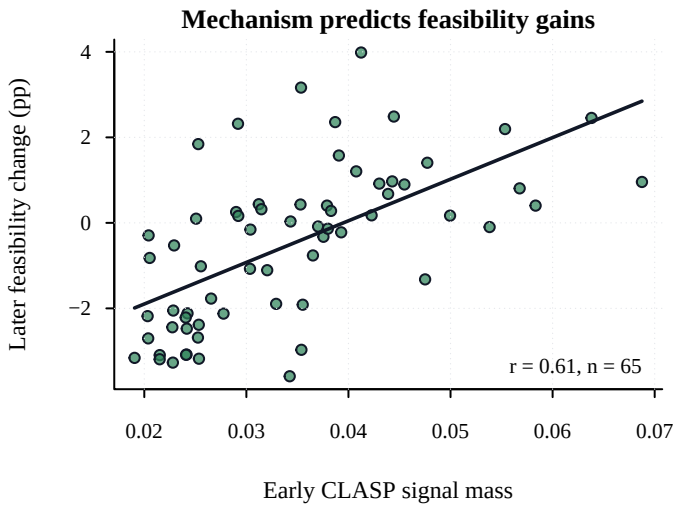

All−infeasible groups occur
All−infeasible groups (%)
0
2
4
6
8
10
0
500
1000
1500
2000
Training step


Scalar reward degenerates
Zero reward−std groups (%)
Scalar GRPO
CLASP
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
Training step


CLASP keeps nonzero coefficients
CLASP signal
advantage L2
alpha
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
Training step


Mechanism predicts feasibility gains
Later feasibility change (pp)
r = 0.61, n = 65
−2
0
2
4
0.02
0.03
0.04
0.05
0.06
0.07
Early CLASP signal mass